StrategyLab report

A Monte Carlo simulation of 2 betting strategies on identical random outcomes. An educational simulation for adults (18+): no real money, not betting advice. US National Problem Gambling Helpline: 1-800-MY-RESET.

Takeaways

- Every bet in this game loses 2.70% of its stake on average (the house edge), so every strategy should lose about 2.70% of everything it wagers; a strategy can change only how much is wagered and how the results are spread out.
- Measured EV per $ wagered, with 95% intervals: Flat -2.76% (-2.84% to -2.68%) and Martingale -2.91% (-3.75% to -2.06%); every interval includes the theoretical -2.70%; the difference between the strategies isn't clear at this sample size.
- Share of sessions that went bust (fell below the table minimum or could not cover the next bet), with 95% intervals: Flat 1.6% (1.4–1.9%) and Martingale 17.1% (16.3–17.8%); Martingale went bust more often than Flat, and their intervals do not overlap.
- Counting busted sessions only, half of Flat's busts came by round 874 and the earliest tenth by round 647.4; half of Martingale's busts came by round 14 and the earliest tenth by round 6.
- Martingale ended ahead in 82.9% (82.2–83.7%) of sessions yet lost money per dollar wagered on average: ending ahead in most sessions is not the same as gaining on average.

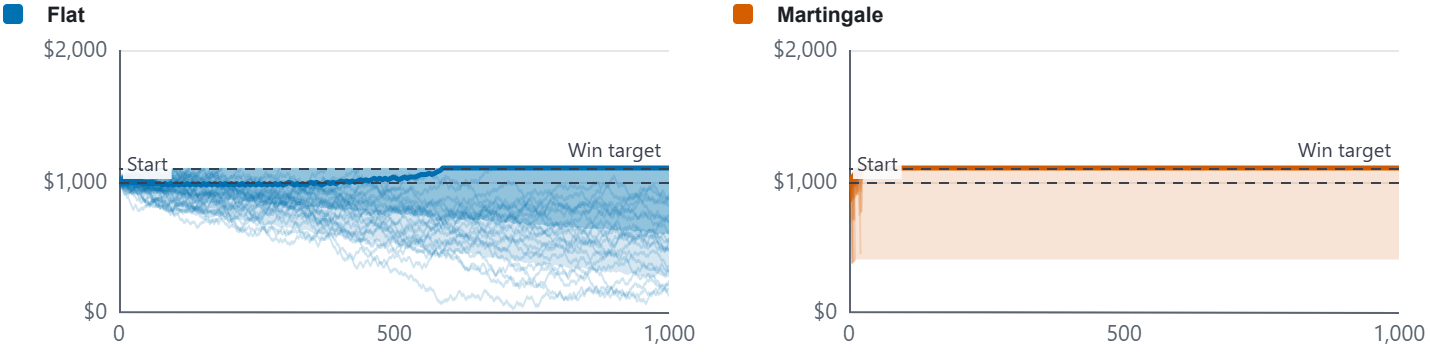
Key statistics

Statistic	Flat	Martingale
EV per $ wagered (95% interval)	-2.76% (-2.84% to -2.68%)	-2.91% (-3.75% to -2.06%)
Theoretical EV per $ (−edge)	-2.70%	-2.70%
P(profit > 0) (95% interval)	53.8% (52.8–54.7%)	82.9% (82.2–83.7%)
P(bust) (95% interval)	1.6% (1.4–1.9%)	17.1% (16.3–17.8%)
Median rounds to bust (busted sessions only)	874	14
p10 rounds to bust (busted sessions only)	647.4	6
P(max drawdown ≥ 50% of start)	30.9%	17.1%
Final bankroll: p5 / median	$240.00 / $1,100.00	$390.00 / $1,100.00
Final bankroll: mean	$842.22	$982.80
Mean total wagered	$5,720.35	$591.25

Intervals are 95%: Wilson intervals for proportions, and EV per $ ± 1.96 standard errors. Rounds to bust count busted sessions only.

Bankroll over time

Middle 90% (light) and 50% (dark) of the first 2,000 sessions, the median (thick line) and 50 sample sessions; every panel uses the same axes.



Configuration

Game: European roulette (even-money); house edge 2.703%
Table limits: minimum $1.00; no maximum
Rounds: at most 1,000 per session; if a bet can't be covered, the session stops
Strategies: Flat (Units per bet: 1); Martingale (Multiplier after a loss: 2)
Bankroll: $1,000.00 to start; base bet $10.00
Stops: win target $1,100.00; no loss floor
Sessions and seed: 10,000 sessions per strategy; master seed 12345

Reproduce this run

Open the link and press Run: it holds the seed and every setting, so the same build gives the same numbers bit for bit. Seed **12345**, 10,000 sessions, build **c81b31b** (github.com/iangopen/strategylab).

https://iangopen.github.io/strategylab/#s=eyJ2Ijo0LCJnIjoiZXVyb3BlYW4iLCJiIjoxMDAwLCJiYiI6MTAsInRuIjoxLCJzdyI6MTEwMCwiciI6MTAwMCwibiI6MTAwMDAsInNkIjoxMjM0NSwic3QiOltbImZsYXQiXSxbIm1hcnRpbmdhbGUiXV19